

MIHIR BANSAL

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Master of Science, Data Science | **GPA: 3.97/4.00**

Aug 2023 - Dec 2024

Courses: Machine Learning, Advanced NLP (Teaching Assistant), Search Engines, Cloud Computing, Multimodal ML

Birla Institute of Technology and Science, Pilani

Hyderabad, India

Bachelor of Science, Computer Science | **GPA: 9.37/10.00**

Aug 2018 - Jul 2022

Courses: Data Structures & Algorithms, Object Oriented Programming, Database Systems, Operating Systems, Computer Networks

TECHNICAL SKILLS

Programming Languages C++, Python, C, Java, C#, R, SQL, MATLAB, Go, Scala, JavaScript, HTML, CSS

Libraries & Frameworks PyTorch, Tensorflow, Keras, NLTK, OpenAI, Transformers, Langchain, Elasticsearch, Databricks, Hive, AWS, GCP, Azure, Spark, Hadoop, Kafka, Kubernetes, Docker, NodeJS, Tableau, CUDA, A/B Testing

WORK EXPERIENCE

ServiceNow

Santa Clara, CA

Senior Machine Learning Engineer

Jan 2025 - Present

- Developed a scalable framework for automated generation of synthetic security evaluation datasets to fine-tune NowLLM for Security Recommendations, AI Search, and Conversational AI assistants on the ServiceNow platform using LLaMA-70B.
- Developed an open-source framework for security evaluation of chat, web-based and computer-use agents, and developed agent safeguards using GPT-4o and LLaMAGuard, improving the Task Success Rate of the Tau-bench conversational AI agent by 24%.
- Assisted the team in hiring by conducting interviews and mentored visiting researchers.

Microsoft

Hyderabad, India

Machine Learning Engineer

Jul 2022 - Aug 2023

- Worked in the Microsoft Search and Recommendations team to improve the XGBoost Machine Learning ranking model for acronym search by adding a multigram matching feature with top user AI graph nodes. Achieved an offline precision (MAP) of 92% and improved the online Click-Through Rate by 24%, monitored via A/B experimentation.
- Developed an end-to-end LLM-based summarizer with webpage summary and FAQs, finetuned using RLHF, for Bing enterprise search using GPT-3 with a scalable query serving microservice and a time-based indexing service using OpenAI and Kubernetes.
- Implemented the data pipeline of a profanity-filtering backend API for Outlook calendar search by using Apache Spark for processing large-scale search logs from a sharded Cassandra database, reducing the search index loading time by 17%.

Seagate Technology

Fremont, CA

Machine Learning Engineer Intern

May 2024 - Aug 2024

- Implemented an auto-merging retrieval approach with PostgreSQL logging in the LLM-based Retrieval Augmented Generation pipeline for a chatbot application with Azure AI Search, improving context relevance by 37% and answer relevance by 32%.

JP Morgan Chase & Co.

Mumbai, India

Quantitative Research Analyst Intern

Jan 2022 - Jun 2022

- Developed a Time Series Rating Migration model to predict credit ratings with macroeconomic scenarios, with Spark-based ETL to process historical ratings data using SQL, and using LSTM and ARIMA models for prediction, achieving an RMSE of 0.19.

University of Hamburg

Hamburg, Germany

Research Intern

Aug 2021 - Dec 2021

- Collaborated with the NLP group on fine-tuning BERT language model with knowledge graph from Wikidata for improving semantics-based question answering using NLTK and Transformers, achieving an improvement of 13% in the GLUE score.

Microsoft

Hyderabad, India

Software Engineer Intern

May 2021 - Jul 2021

- Improved the coverage of Outlook Calendar Search by 37% by performing optimized REST API calls and caching to index grouped user entities for person-based Natural Language queries using Elasticsearch and Redis, reducing the overall response time by 60.8%.

PUBLICATIONS

1. Leo Boisvert*, **Mihir Bansal***, Chandra Kiran Reddy Evuru*, Gabriel Huang*, Abhay Puri*, Avinandan Bose*, Maryam Fazel, Quentin Cappart, Jason Stanley, Alexandre Lacoste, Alexandre Drouin, Krishnamurthy Dvijotham. *DoomArena: A framework for Testing AI Agents Against Evolving Security Threats*. Conference on Language Modeling. 2025. [Paper]

*equal contribution and joint primary authorship

2. Meriem Beloucif, **Mihir Bansal**, Chris Biemann. *Using Wikidata for Enhancing Compositionality in Pretrained Language Models*. Recent Advances in Natural Language Processing. 2023. ACL Anthology. [Paper]

3. B.S.A.S. Rajita, **Mihir Bansal**, Bipin Sai Narwa, Subhrakanta Panda. *Cuckoo search in threshold optimization for better event detection in social networks*. Social Network Analysis and Mining. 2022. Springer. [\[Paper\]](#)

PROJECTS

- Multimodal Retrieval Augmented Generation for Conversational Visual QA** | *CMU* *Aug 2024 - Dec 2024*
- Implemented Multimodal Retrieval Augmented Generation with CLIP encoder and GPT-4, and LLM-based conversational history summarization, improving F-1 score of Multimodal Conversational VQA on the MMConvQA dataset by 57%.
- De-biasing LLMs with Retrieval Augmented Generation** | *Research Assistant, CMU* *Jan 2024 - Dec 2024*
- Worked with Prof.Graham Neubig on developing a framework for mitigating bias in pre-trained Mistral-7b and LLaMA-7b LLMs using Retrieval Augmented Generation for generating synthetic data from the 2020 American Elections survey, achieving an improvement of 21% in average stance score.
- Improving Search Engines with LLM Embeddings** | *Research Assistant, CMU* *Jan 2024 - May 2024*
- Worked with Prof.Jamie Callan on improving Neural Information Retrieval models by using FAISS and Hypothetical Document Embeddings from GPT-3.5, improving the MAP by 17%, NDCG by 29% and Recall by 18% on the MSMARCO dataset.
- Twitter Recommendations System with Apache Spark** | *CMU* *Jan 2024 - May 2024*
- Developed a highly scalable end-to-end Twitter topics recommendations service by ranking user topics for 10M users by their PageRank scores in a large Twitter social network graph with Azure Databricks, Scala and Spark SQL.
- Cab Booking Application with Apache Kafka** | *CMU* *Jan 2024 - May 2024*
- Implemented a real-time distributed stream processing cab driver-user matching backend service with an ads recommendations and pricing microservice from user clicks by using Apache Kafka, Samza and Amazon Elastic MapReduce.
- Adapting Multilingual Visual LLMs for Code-mixed Visual QA** | *CMU* *Aug 2023 - Dec 2023*
- Finetuned the mBLIP LLM model for low-resource Hinglish code-mixed VQA by using LoRA and QLoRA finetuning, achieving an improvement of 34% in accuracy on the MS-COCO dataset as compared to the baseline model.